

Standard Form & Indices

Test 2 · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $3^6 \times 3^2$ as a single power of 3.

[1 mark]

A2. Write $8^7 \div 8^3$ as a single power of 8.

[1 mark]

A3. Write $(5^3)^2$ as a single power of 5.

[1 mark]

A4. Work out the value of $3^2 \times 3^3$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 9^0 .

[1 mark]

B2. Work out the value of 4^{-2} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 10^{-4} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 54,000 in standard form.

[2 marks]

C2. Write 6,200,000 in standard form.

[2 marks]

C3. Write 0.0067 in standard form.

[2 marks]

C4. Write 0.00009 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 4.2×10^3 as an ordinary number.

[1 mark]

D2. Write 7×10^{-4} as an ordinary number.

[1 mark]

D3. Work out $(3 \times 10^2) \times (3 \times 10^4)$. Give your answer in standard form.

[3 marks]

D4. Work out $(8 \times 10^7) \div (2 \times 10^2)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 2^6 .

[1 mark]

E2. $\sqrt{90}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $p^5 \times p^4$.

[1 mark]

F2. Simplify $q^{11} \div q^4$.

[1 mark]

F3. Simplify $(r^2)^5$.

[1 mark]

F4. Simplify $4s^3 \times 3s^4$.

[2 marks]

F5. Simplify $(2t^3)^4$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $36^{1/2}$.

[1 mark]

G2. Work out the value of $8^{1/3}$.

[1 mark]

G3. Work out the value of $81^{3/4}$.

[2 marks]

G4. Work out the value of $16^{-1/2}$.

[2 marks]